

Salpeter equation with complex Hamiltonian

Fernando Alonso-Marroquin*

CIPR, King Fahd University of Petroleum and Minerals, Dhahran, 31261, Kingdom of Saudi Arabia

Yaoyue Tang

Modelling and Simulation Research Group, The University of Sydney, Sydney NSW 2006, Australia

Fatemeh Gharari 

Department of Statistics and Computer Science, University of Mohaghegh Ardabili, Ardabil, Iran

M. N. Najafi[†]

Department of Physics, University of Mohaghegh Ardabili, P. O. Box 179, Ardabil, Iran



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We use integral representations and analytical solutions for the propagator of the $(1 + 1)$ -dimensional Salpeter Hamiltonian to describe the wave propagation of a non-spin-relativistic quantum particle. An explicit expression of the propagator, or Green function, is derived in terms of special functions. The exact Green function is convoluted with the initial wave function by using suitable boundary conditions to find the wave propagation for both massive and massless particles. The analytical extension of the Hamiltonian in the complex plane allows us to formulate the equivalent stochastic problem, namely the Bäumers equation. This equation describes *relativistic* stochastic processes with time-changing anomalous diffusion. Given the known crossover between Cauchy and Gaussian diffusion, we analytically calculate the propagator to interpolate between these regimes. This Bäumers propagator corresponds to the Green function of a relativistic diffusion process that interpolates between Cauchy distributions for small times and Gaussian diffusion for large times, providing a framework for stochastic processes where anomalous diffusion is time-dependent.

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I. INTRODUCTION

The importance of relativistic quantum mechanics (RQM) extends beyond its role in the standard model of elementary particles [1]. RQM is also a crucial framework for describing the new phases of matter, such as two-dimensional electron gas [2], graphene [3], relativistic plasmas [4–6], neutron stars [7], Weyl semimetals [8], relativistic hydrodynamics [9], and other materials with Dirac cone dispersion in their band structures [10]. RQM has broad applications in other fields, such as diffusion and stochastic phenomena, where it is implemented through a Wick rotation in time. Examples include relativistic stochastic processes [11–14], the relativistic Ornstein-Uhlenbeck process [15,16], relativistic Brownian motion [17,18], the relativistic Fokker-Planck equation [19–21], the normal and relativistic Gross-Pitaevskii equation [22], and relativistic diffusion [23]. Given its extensive applications, the standard theory for relativistic (both classical and quantum) statistical mechanics is now well developed [24]. Various approaches exist to construct a relativistic version of traditional quantum mechanics that adheres to the principles of relativity through a covariant formulation.

These approaches include the Klein-Gordon equation [1], characterized by a second-order time derivative, and the Dirac equation [25], with its first-order time derivative. The Salpeter equation [26–28], derived from the Salpeter Hamiltonian, is used in relativistic Hamiltonian quantum mechanics to introduce the probability current for a quantum spinless relativistic particle. It serves as an alternative to the Klein-Gordon equation, and it is often referred to as the *square root* of the Klein-Gordon operator. All three of these are well-known formulations in this context. The Bäumers Hamiltonian is derived from the analytical extension of the Salpeter Hamiltonian in the complex plane. This extension allows for the formulation of an equivalent stochastic problem. The Salpeter equation is a relativistic extension of the Schrödinger equation designed to describe bound states in quantum field theory [29], particularly for two-particle systems [30]. As a reduced form of the Bethe-Salpeter equation, it simplifies the interaction between relativistic particles by incorporating the instantaneous approximation of the interaction kernel [31]. The equation has been widely used in the study of relativistic composite particles described as quantum bound states, such as in meson spectroscopy [32], and it has played a critical role in understanding quantum chromodynamics [33]. In this context, the Salpeter equation addresses relativistic corrections to nonrelativistic quantum systems, particularly focusing on the wave functions of bound states and their propagators,

*Contact author: fernando.marroquin@kfupm.edu.sa

[†]Contact author: morteza.nattagh@gmail.com

making it a key tool for high-energy particle physics. Despite its simplifications, it retains crucial relativistic effects, including time dilation and relativistic mass corrections, thus providing a more accurate framework for modeling relativistic quantum systems compared to nonrelativistic approaches. The Salpeter equation, in its simplest form, governs scalar particles through its relativistic covariance, applied at the operator level, while the Dirac and Klein-Gordon equations describe higher-dimensional representations of the Lorentz group. Although working with a square root of the Klein-Gordon operator makes the Salpeter equation more challenging to handle, it resolves critical issues in the Klein-Gordon equation, such as the problem of negative energy particles. This advantage makes the Salpeter equation a subject of intense study [28]. However, the square-root operator introduces non-locality into Salpeter's equation, adding further complexity. An analysis of the conserved current, along with a second quantization formulation, can be found in [34]. Despite its significance, there is limited research focused on the structure and implications of the equation. For example, the causality structure of this model, its plane-wave solutions, the superposition of quantum states, the entanglement structure, the second-quantized field theory description, particle algebra that restores causality, the dual stochastic theory (after Wick rotation), and its closed form of the Green function are under-explored in the literature.

A key motivation for the stochastic interpretation of the Salpeter equation arises from the crossover phenomenon in diffusion processes. This crossover behavior occurs, for example, in the analysis of stochastic processes, such as the evolution of the probability density function of the S&P 500 stock market index [35]. This crossover manifests itself as a superdiffusive regime at short times, transitioning to a Gaussian diffusive regime at larger times. Attempts have been made to use quantum-mechanical models to measure the volatility of the stock return, with studies reporting a power-law relation in the tails of price-return distributions [36]. At small momentum values, the Salpeter equation behaves analogously to the Schrödinger equation, where in the dual stochastic equation, the solution describes normal diffusion. At large momentum values, it corresponds to massless free particles [28]. Moreover, Green's function of the stochastic equation is related to Cauchy distributions [37]. Therefore, we expect a crossover between these two limits for intermediate momenta, which induces a corresponding crossover in spatiotemporal coordinates.

In quantum mechanics, both analytical and numerical solutions are needed because while analytical solutions provide exact, rigorous results for simple systems, they are often unavailable for complex systems, requiring numerical methods for approximations. For example, the Schrödinger equation is generally solved numerically, although its analytical solutions are known.

This paper derives and validates solutions using complex variable analysis and rigorous numerical testing. Transforming integrals in the complex plane yields stable numerical solutions that precisely match analytical results. These techniques can be extended to solve more complex problems lacking analytical solutions. Complex analysis enables the formulation of the Salpeter equation as an integral representation,

leading to a solution for the free-particle propagator and its associated stochastic interpretation.

First, the strong oscillations of the integrand are removed by transforming the integral in the complex plane. Then, the inherent singularities of the Salpeter propagator are circumvented to obtain a solution for wave propagation that attains the conservation of probability. The solutions of the Salpeter propagator and wave functions are not unique, since several solutions can be obtained depending on how the singularities are circumvented. However, we achieved consistency by demonstrating that the numerical solution of its integral representation matches perfectly with the analytical solution. Beyond calculating the Green function, we also determine the Salpeter wave function and connect it to a generalized form of the Bäumler equation, which has applications in stochastic processes with slow convergence to Gaussian dynamics such as stock markets. We examine several asymptotic solutions—confirmed numerically—using the general properties of the integral representation for both the Salpeter and Bäumler equations. Through scaling arguments, we investigate the internal structure of the model in Appendix B. A perturbative approach yields an expansion for the Green function, applicable in asymptotic limits; see Appendix E. We also provide an analysis of the solutions to the Salpeter differential equation (Appendix F). The inherent singularities of the quantum propagator are removed by imposing appropriate boundary conditions on the initial wave function. In the stochastic formulation, the singularities disappear when a Wick rotation is applied, resulting in a closed form for the Green function of the relativistic diffusion equation.

The body of this paper is organized as follows. In the next section, we introduce the Salpeter equation to be analyzed, providing the integral representation and analytical solutions of the propagator. In Sec. III, we present the dual stochastic description, namely the Bäumler equation, and we explore the propagator's crossover regimes for this relativistic stochastic process.

II. SALPETER HAMILTONIAN

Our derivation is based on the traditional formulation of the RQM of a free particle, resulting in an integral representation of the Green function (propagator); see Appendix A. The propagator function $G(x, t)$ is used to calculate the wave function $\psi(x, t)$ in terms of its initial value $\psi_0(x)$ as

$$\psi(x, t) = \int_{-\infty}^{\infty} G(x - x_0, t) \psi_0(x_0) dx_0. \quad (1)$$

The propagator depends on the Hamiltonian of the particle and is derived in Appendix A using textbook quantum mechanics:

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}[H(p)t + px]} dp. \quad (2)$$

Let us assume that the Hamiltonian is an even function of the momentum $H(-p) = H(p)$. There are two consequences from this property: First, the propagator will be an even function with respect to x , i.e., $G(-x, t) = G(x, t)$. Second,

the propagator can be calculated as the sum $G = G_+ + G_-$ as

$$G_{\pm}(x, t) = \frac{1}{2\pi} \int_0^{\infty} e^{-\frac{i}{\hbar} H(p)t \pm \frac{i}{\hbar} px} dp. \quad (3)$$

Adding $G_+(x, t)$ and $G_-(x, t)$, we obtain a useful expression of the propagator $G(x, t)$ as

$$\begin{aligned} G(x, t) &= G_+(x, t) + G_-(x, t) \\ &= \frac{1}{2\pi} \int_0^{\infty} e^{-\frac{i}{\hbar} H(p)t} (e^{\frac{i}{\hbar} px} + e^{-\frac{i}{\hbar} px}) dp \\ &= \frac{1}{\pi} \int_0^{\infty} e^{-\frac{i}{\hbar} H(p)t} \cos\left(\frac{px}{\hbar}\right) dp. \end{aligned} \quad (4)$$

Let us consider the Hamiltonian of a free, massive, and spinless relativistic particle in one dimension, namely the Salpeter Hamiltonian,

$$H_S(p) = \sqrt{m^2 c^4 + c^2 p^2} - mc^2. \quad (5)$$

Due to the square root, there is no standard differential equation for this Hamiltonian. In Eq. (5), we subtract the rest of the energy mc^2 to recover the traditional Schrödinger equation in the limit of small velocities $|p| \ll mc$. This limit leads to

$$H_S(p) \approx \frac{p^2}{2m}. \quad (6)$$

Then the traditional Schrödinger equation can be expressed in space coordinates as a partial differential equation

$$i\partial_t \psi(x, t) = -\frac{\hbar^2}{2m} \partial_x^2 \psi(x, t). \quad (7)$$

A simple transformation of the Schrödinger equation in quantum mechanics can lead to the diffusion equation in a stochastic equation. This transformation can be performed by either transforming the time into imaginary ($t \rightarrow it$) or transforming the Hamiltonian into imaginary ($H \rightarrow iH$). Either transformation leads to normal diffusion in the stochastic process,

$$\partial_t P(x, t) = D \partial_x^2 P(x, t), \quad (8)$$

where $D = \hbar^2/(2m)$. In the relativistic case, the Schrödinger equation is transformed to a form in which the order of the time and the spatial derivatives are the same. In this case, the structure of the propagator and its implications on wave propagation and relativistic diffusion are not well-known. Dirac provided a relativistic equation by converting the wave equation into a four-dimensional function that inherits the spin of the particles. An alternative approach is to assume that the wave function is scalar, representing a relativistic quantum particle without spin. As demonstrated in Appendix A, the quantum bra-ket (bracket) formalism allows us to express the propagator as integral, given by Eq. (2), without relying on fractional differential equations. Instead, our analysis is based on performing an analytical continuation of the Salpeter Hamiltonian in the complex plane. First, the Hamiltonian is written as

$$H_S(p) = mc^2 \left[f\left(\frac{p}{mc}\right) - 1 \right], \quad (9)$$

where $f(p) = \sqrt{1 + p^2}$. Then we define $f(z)$, where $z = p + iq$, as the analytical continuation of $f(p)$. Figure 1 shows the real and imaginary parts of the principal analytical branch of the complex function $f(z) = \sqrt{1 + z^2}$. The real part of the function has a saddle point at $z = 0$. The function is analytic in some regions that contain the real axis $\text{Im}(z) = 0$. This is where the real Hamiltonian function is defined. The complex function is analytical in the complex plane except for the two branch cuts $[-i\infty, -i]$ and $[i, i\infty]$, where the derivatives do not exist. We can slightly modify this function by using the function $f^*(p + iq) = \lim_{\epsilon \rightarrow 0} f(p + q + i\epsilon)$, where $\epsilon > 0$. This will correct the values along the imaginary axis so that the new function becomes analytical in the first and fourth quadrants, including the imaginary axis.

For simplicity, in the rest of this paper, we rescale the variables x , t , and p so that $\hbar = c = 1$. Thus Eq. (5) becomes

$$H_S(p) = \sqrt{m^2 + p^2} - m. \quad (10)$$

Then substituting Eq. (10) into Eq. (4), the propagator we want to solve can be defined as

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iH_S(p)t + ipx} dp. \quad (11)$$

We split the Green function as $G = G_+ + G_-$, where

$$G_{\pm}(x, t) = \frac{1}{2\pi} \int_0^{\infty} e^{-iH_S(p)t \pm ipx} dp. \quad (12)$$

In an earlier paper by Lämmerzhal, an analytical derivation of this propagator was proposed in terms of special functions [34]. However, these solutions were conditioned by singularities in the light cone. On the other hand, the numerical solution to the above integrals is highly unstable due to the strong oscillatory behavior of the integrands. We will demonstrate that more stable numerical solutions can be achieved by using complex variable analysis to derive the integral representation of the propagators. In addition, we will provide analytical solutions for these integral representations. In the following sections, we present stable integral representations and analytical solutions for different cases. We will consider the analytical extension of the integrands and employ complex variable analysis to handle the integrals.

A. Case $m > 0$ and $t \gg 1$ ¹

We begin with the integration of Eq. (11) for the case in which t is large. First, we perform an analytical continuation for the integrand in Eq. (11) in the complex plane. In particular, the integrand is an analytical function in a region of the complex plane that includes the real axis ($z = p + iq$ and $q = 0$). We consider t to be real and positive. When t is large, the primary contribution to the integrand occurs when $H_S(p)$ is imaginary and minimal, which corresponds to $p = 0$. Under these conditions, we can apply Debye's method based on the *saddle point approximation*, as outlined in Section 31 of the book by Dennery and Krzywicki [38]. This method involves

¹In dimensional form, the case is $mtc^2/\hbar \gg 1$, so that this limit is valid for particles with relatively large mass or relatively large times.

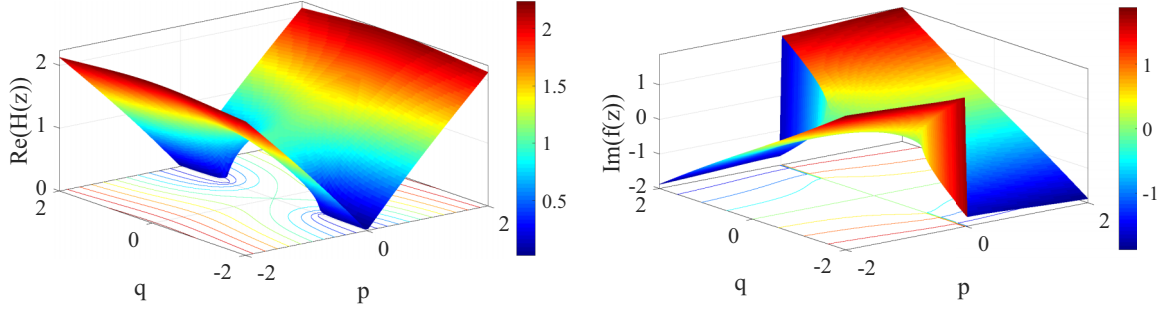


FIG. 1. Real and imaginary parts of the function $f(z) = \sqrt{1+z^2}$, where $z = p + iq$. The principal branch used in this calculation is obtained from the branch of the complex square root: $\sqrt{z} = \sqrt{\rho}(\cos(\phi/2) + i \sin(\phi/2))$, where $z = \rho e^{i\phi}$ and $-\pi \leq \phi \leq \pi$. This branch is extended along the imaginary axis by imposing $\text{Im}f(iq) > 0$ in the first quadrant and $\text{Im}f(iq) < 0$ in the fourth quadrant. This guarantees that the defined function is analytical in the first and fourth quadrants of the complex plane.

expanding the Hamiltonian around its minimum point $p = 0$ up to the quadratic term, that is, $H_S(p) \approx p^2/(2m)$. Thus,

$$G(x, t) \sim \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\frac{p^2}{2m}t + ipx} dp = \sqrt{\frac{m}{2\pi it}} e^{-\frac{mx^2}{2it}}. \quad (13)$$

Here, we use the following integration formula:

$$\int_{-\infty}^{\infty} e^{iap - bp^2} dp = \sqrt{\frac{\pi}{b}} e^{\frac{a^2}{4b}}.$$

Note that this solution corresponds to the propagator of traditional quantum mechanics. The evolution of a wave function can be obtained from its initial condition using Eq. (A17). We can also recover the Green's function of the diffusion equation by transforming $t \rightarrow it$ and $1/(2m) \rightarrow D$:

$$G(x, t) \rightarrow \frac{1}{\sqrt{\pi Dt}} e^{-\frac{x^2}{4Dt}}. \quad (14)$$

The propagator accounts for the probability distribution function of the particle at $x = 0$ and $t = 0$, which corresponds to the classical diffusion process.

B. Integral representation, case $m > 0$ and $0 < x < t$

In this section, we derive an integral representation of the propagator, where the integrand does not exhibit strong oscillatory behavior. The method is based on the Cauchy theorem from complex variable analysis, which is used to convert the integral on the real axis into an integral over the imaginary axis, effectively eliminating heavy oscillations. As a result, the transformed integral becomes well defined and easily integrable. The transformation of an integral from the real axis to the imaginary axis relies on two key conditions: first, the integrand must be analytically within a quadrant of the complex plane; and second, the integrand must vanish as the complex argument becomes sufficiently large in that quadrant. The detailed use of complex variable analysis is provided in two theorems, formulated and proven in Appendix C. Here, we demonstrate that more stable numerical solutions can be achieved by applying these theorems to obtain an integral representation of the propagators. In addition, we provide analytical solutions for these integral representations.

First, we present an integral representation for the propagator of relativistic particles with arbitrary mass $m > 0$ inside the light cone ($0 < x < t$). We substitute Eq. (10) into Eq. (12)

to obtain

$$G_{\pm}(x, t) = \frac{1}{2\pi} \int_0^{\infty} e^{\Phi_{\pm}(p)} dp, \quad (15)$$

where

$$\Phi_{\pm}(p) = -i\sqrt{m^2 + p^2}t - mt \pm ipx. \quad (16)$$

Now we consider that for $0 < x < t$ and $-\pi/2 < \theta \leq 0$, the condition

$$\lim_{R \rightarrow \infty} \text{Re}\Phi(Re^{i\theta}) = \lim_{R \rightarrow \infty} -R(-t \pm x) \sin(\theta) = -\infty. \quad (17)$$

This condition is satisfied in the fourth quadrant in the complex plane. Thus, applying Theorem 2 (see Appendix C) to the analytical branch shown in Fig. 1 and substituting p by $-iq$, we have

$$\begin{aligned} G_{\pm}(x, t) &= \frac{e^{imt}}{2\pi} \int_0^{\infty} e^{-i\sqrt{m^2 + p^2}t \pm ipx} dp \\ &= \frac{e^{imt}}{2\pi} \int_0^{\infty} e^{-i\sqrt{m^2 + (-iq)^2}t \pm i(-iq)x} d(-iq) \\ &= \frac{e^{imt}}{2i\pi} \left(\int_0^m + \int_m^{\infty} \right) e^{-i\sqrt{m^2 + (-iq)^2}t \pm iqx} dq \\ &= \frac{e^{imt}}{2i\pi} \left(\int_0^m e^{-i\sqrt{m^2 - q^2}t \pm iqx} dq + \int_m^{\infty} e^{-\sqrt{q^2 - m^2}t \pm iqx} dq \right). \end{aligned} \quad (18)$$

The full propagator $G(x, t) = G_+(x, t) + G_-(x, t)$ becomes

$$G(x, t) = \frac{e^{imt}}{i\pi} \int_0^{\infty} e^{-\sqrt{q^2 - m^2}t} \cosh(qx) dq. \quad (19)$$

The integrand in this equation does not have singular oscillations, and it is easily integrated by standard numerical methods.

C. Integral representation, case $m > 0$ and $x > t > 0$

Now we obtain the expression for the propagator outside of the light cone ($x > t > 0$). Under these conditions, Φ_- in Eq. (16) satisfies the condition Eq. (17) in the fourth quadrant where $\sin \theta < 0$. For Φ_+ we note that $-t + x > 0$. In this case, the integral is calculated in the first quadrant, where $\sin \theta > 0$. Therefore, we only apply the loop integral in the

first quadrant for Φ_+ , and in the fourth quadrant for Φ_- to Eq. (12). Applying Theorem 2 on G_- for the first quadrant and using the change of variable $p \rightarrow iq$,

$$\begin{aligned} G_-(x, t) &= \frac{e^{imt}}{2\pi} \int_0^\infty e^{-i\sqrt{m^2+p^2}t-ipx} dp \\ &= \frac{e^{imt}}{2\pi} \int_0^\infty e^{-i\sqrt{m^2+(-iq)^2}t-i(-iq)x} d(-iq) \\ &= \frac{e^{imt}}{2i\pi} \left(\int_0^m e^{-i\sqrt{m^2-q^2}t-qx} dq + \int_m^\infty e^{-\sqrt{q^2-m^2}t-qx} dq \right). \end{aligned} \quad (20)$$

And applying Theorem C1 on G_+ for the fourth quadrant and using the change of variable $p \rightarrow -iq$,

$$\begin{aligned} G_+(x, t) &= \frac{e^{imt}}{2\pi} \int_0^\infty e^{-i\sqrt{m^2+p^2}t+ipx} dp \\ &= \frac{e^{imt}}{2\pi} \int_0^\infty e^{-i\sqrt{m^2+(iq)^2}t+i(iq)x} d(iq) \\ &= \frac{-e^{imt}}{2i\pi} \left(\int_0^m e^{-i\sqrt{m^2-q^2}t-qx} dq + \int_m^\infty e^{\sqrt{q^2-m^2}t-qx} dq \right). \end{aligned} \quad (21)$$

Adding both parts and using the identity $1/i = -i$, the full propagator becomes

$$G(x, t) = \frac{ie^{imt}}{\pi} \int_m^\infty \sinh(\sqrt{q^2 - m^2}t) e^{-qx} dq. \quad (22)$$

This solution corresponds to a nonzero propagator outside the light cone, exhibiting exponential decay. However, this faster-than-light propagation does not violate causality, since the key question in causality is whether a measurement at one point can influence a measurement at another point separated by a spacelike interval. To address this, one must compute the commutator of the fields, which vanishes outside the light cone [39]. A more detailed discussion of causality can be found in quantum field theory, where particles are treated as excitations of the vacuum state or quantum fields. In this context, the Green function is expressed in terms of expectation values of quantum fields, with commutation (or anticommutation) relations applied to bosonic and fermionic creation and annihilation operators [39].

D. Massless particles: Case $m = 0$

When the particle has no mass, the propagator for $0 < x < t$ is calculated by taking $m = 0$ in the last line of Eq. (18),

$$G_\pm(x, t) = \frac{1}{2i\pi} \int_0^\infty e^{-qt \pm qx} dq = \frac{1}{2i\pi} \frac{-1}{-t \pm x}. \quad (23)$$

Adding both parts, the full propagator of massless particles for $x < t$ becomes

$$G(x, t) = \frac{1}{\pi} \frac{it}{x^2 - t^2}. \quad (24)$$

For the case $0 < t < x$, the propagator is obtained taking

$m = 0$ in Eqs. (20) and (21),

$$\begin{aligned} G_-(x, t) &= \frac{1}{2i\pi} \int_0^\infty e^{-qt - qx} dq = \frac{1}{2i\pi} \frac{1}{x + t}, \\ G_+(x, t) &= \frac{i}{2\pi} \int_0^\infty e^{qt - qx} dq = \frac{i}{2\pi} \frac{1}{x - t}. \end{aligned} \quad (25)$$

Adding both parts, the full propagator of massless particles for $x > t$ becomes

$$G(x, t) = \frac{1}{\pi} \frac{it}{x^2 - t^2}. \quad (26)$$

This result is consistent with Eq. (24) for the case in which $x < t$. We notice a singularity at the light cone $x = t$ and, again, nonzero values at $x > t > 0$, indicating faster-than-light propagation. As with massive particles, a quantum field theory formulation is required to avoid the apparent violation of the causality principle in the traditional interpretation of quantum mechanics.

E. Analytical solution

In this subsection, we establish the analytical solution for the Salpeter propagator. Substituting the Salpeter Hamiltonian [Eq. (10)] into the propagator [Eq. (4)], we have

$$\begin{aligned} G(x, t) &= \frac{1}{\pi} \int_0^\infty e^{-i(\sqrt{m^2+p^2}-m)t} \cos(px) dp \\ &= \frac{1}{\pi} e^{imt} \int_0^\infty e^{-i\sqrt{m^2+p^2}t} \cos(px) dp. \end{aligned} \quad (27)$$

Using the identity [28]

$$\begin{aligned} \int_0^\infty dx \exp(-\alpha\sqrt{x^2 + \beta^2}) \cos\gamma x \\ = \frac{\alpha\beta}{\sqrt{\alpha^2 + \gamma^2}} K_1(\beta\sqrt{\alpha^2 + \gamma^2}), \quad \text{Re}[\alpha] > 0, \text{Re}[\beta] > 0, \end{aligned} \quad (28)$$

we can rewrite the propagator as

$$G(x, t) = \frac{1}{\pi} e^{imt} \int_0^\infty e^{-\alpha\sqrt{\beta^2+p^2}} \cos(\gamma p) dp, \quad (29)$$

where $\alpha = it$, $\beta = m$, and $\gamma = x$. Then according to the right-hand side of Eq. (28), we have

$$G(x, t) = \frac{imt e^{imt}}{\pi \sqrt{x^2 - t^2}} K_1(m\sqrt{x^2 - t^2}), \quad x^2 > t^2, \quad (30)$$

where K_1 is the modified Bessel function of the second kind. For $x^2 < t^2$, we can utilize the property of the modified Bessel functions, given as

$$K_\alpha(x) = \frac{\pi}{2} i^{\alpha+1} H_\alpha^{(1)}(ix), \quad -\pi < \arg x \leq \frac{\pi}{2}, \quad (31)$$

where $H_\alpha^{(1)}$ is the Hankel function of the first kind. Replacing $x^2 - t^2$ by $-(t^2 - x^2)$ in Eq. (30), we have

$$\begin{aligned} G(x, t) &= \frac{imt e^{imt}}{\pi \sqrt{-(t^2 - x^2)}} \frac{\pi}{2} i^2 H_1^{(1)}(im\sqrt{-(t^2 - x^2)}) \\ &= -\frac{mt e^{imt}}{2\sqrt{t^2 - x^2}} H_1^{(1)}(-m\sqrt{t^2 - x^2}), \quad x^2 < t^2. \end{aligned} \quad (32)$$

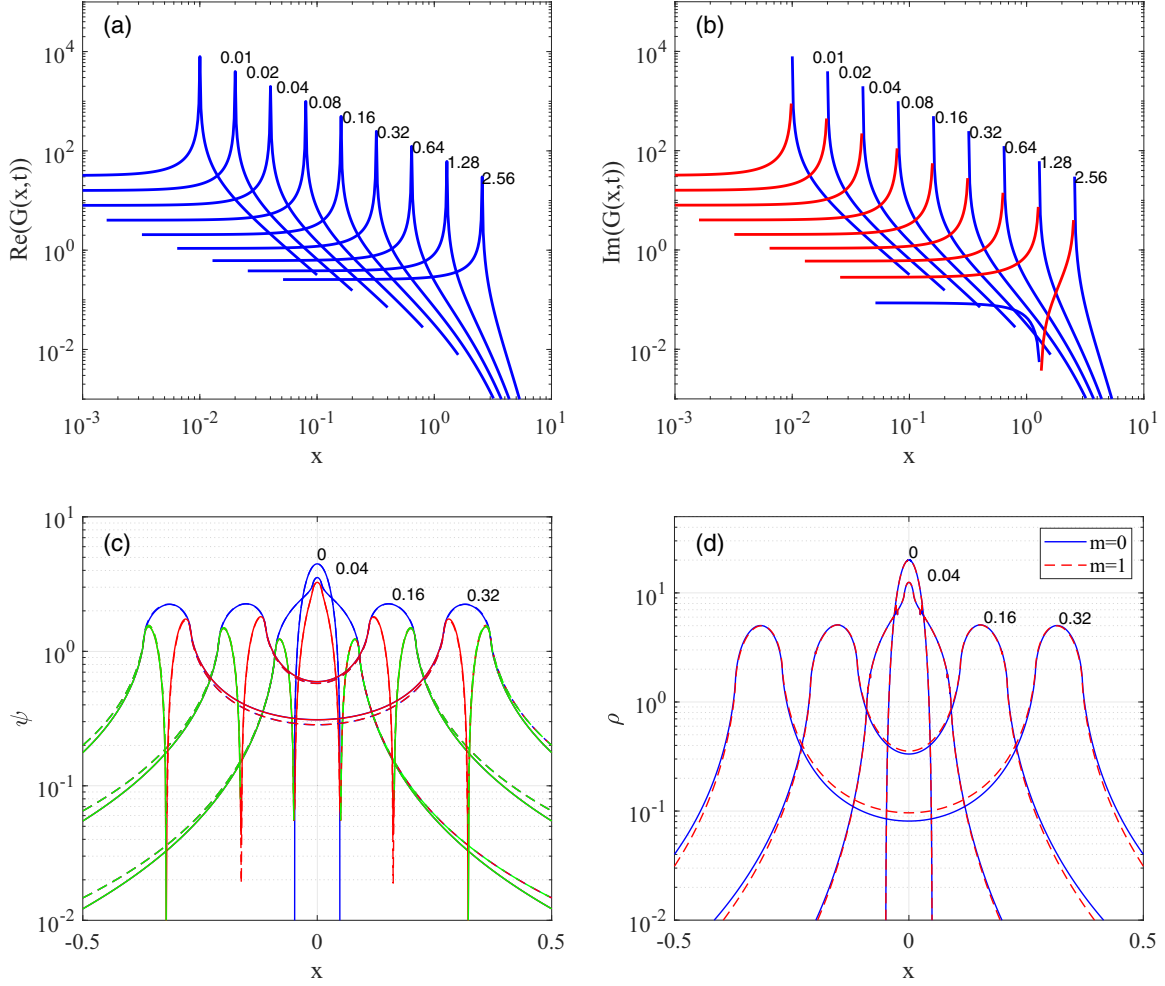


FIG. 2. (a) Comparison between the numerical and analytical solution of the Salpeter propagator $G(x, t)$ concerning x , across various t values. Part (a) shows the real part, and (b) shows the positive (blue) and negative (red) values of the imaginary parts in the log-log scale. The results from both methods exhibit very good agreement. Notably, the propagator displays a singularity at the light cone, represented by $x = t$. (b) The phase shows an abrupt change of $-\pi/2$ at the light cone in both numerical and analytical solutions. Part (c) shows the wave function for massive particles calculated using Eq. (36). The real part is plotted in blue, and the imaginary part is plotted in green for positive values and red for negative values. (d) Probability density for a massless particle $m = 0$ and massive particles $m = 1$.

In summary, the analytical solution of the Salpeter propagator is

$$G(x, t) = \begin{cases} \frac{imt e^{imt}}{\pi \sqrt{x^2 - t^2}} K_1[m \sqrt{x^2 - t^2}], & x^2 > t^2, \\ -\frac{mt e^{imt}}{2\sqrt{t^2 - x^2}} H_1^{(1)}[-m \sqrt{t^2 - x^2}], & x^2 < t^2, \end{cases} \quad (33)$$

where $H_1^{(1)}$ is the Hankel function of the first kind, and K_1 is the modified Bessel function of the second kind. This solution is consistent with the analytical solution by Lämmerzahl [34]. The derivation proposed by Lämmerzahl is a nonexplicit expression in terms of Hankel functions with nonunique solutions. Actually, the solution in this perspective will depend on how the singularity is circumvented. Since our solution is explicit, it effectively removes the singularity and provides a physically sound solution, as will be shown in the rest of the paper.

In Fig. 2, we present numerical and analytical solutions for the real and imaginary parts of the propagator and the phase at varying t . Both solutions are indistinguishable. We

also compare the cases in which $m = 1$ and 0 . The numerical solution inside the light cone $x < t$ is obtained by numerical integration of Eq. (19). The numerical solution outside the cone $x > t$ is the numerical integration of Eq. (22). We observe that the hyperbolic functions of these integrals are sensitive to large arguments. However, this issue can be solved by expressing them in terms of exponential functions. The analytical solution is calculated by substituting the corresponding values of x and t into Eq. (33). In this context, we choose $t = 2^n/100$, $n = 0, 1, 2, \dots, 8$, and $x \in [0, 5t]$ as the calculation range. The solutions for $m = 1$ and 0 are asymptotically similar when $d = |x - t| \rightarrow 0$. Here d is the distance to the light cone $|x| = t$. For each time t , the propagators present a singularity at $d = 0$. This singularity proves to be of the order $1/d$, so it produces a sudden change in the imaginary part of the propagator and a discontinuity of the argument of this function. As will be discussed in the next section, this singularity requires careful treatment when calculating the wave function of the Salpeter equation.

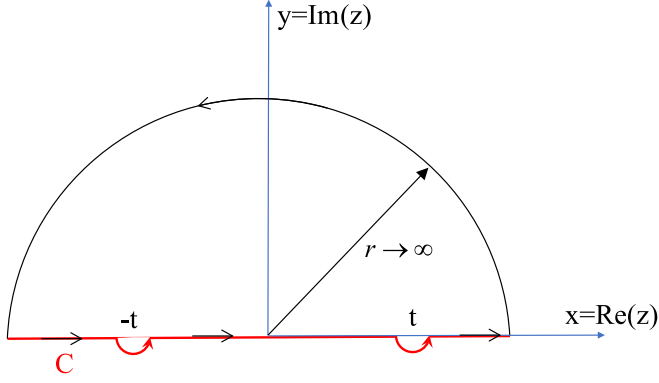


FIG. 3. Contour in the complex plane used to calculate the Salpeter wave function.

F. Salpeter wave function

Using Eq. (1), one finds the wave function in later times, given its form in the initial time. After the change of variable $x' = x - x_0$, this equation can be written as

$$\psi(t, x) = \int_{-\infty}^{\infty} G(x', t) \psi_0(x - x') dx'. \quad (34)$$

In a strict sense, this integral does not exist because the propagator has two first-order singularities at $x = \pm t$. To obtain a meaningful solution for the wave function, we need to circumvent the singularities by choosing an appropriate integration path in the complex plane. A meaningful solution is obtained if the integration is performed on the deformed path C shown in Fig. 3. We redefine the wave function as the path integral

$$\psi^*(t, x) = \int_C G(z, t) \psi_0(x - z) dz. \quad (35)$$

This integral can be performed by applying the theorem of residues of the complex variable [38] over the closed path shown in Fig. 3. After calculating the residues inside the closed path, the integration becomes

$$\begin{aligned} \psi^*(t, x) = P \int_{-\infty}^{\infty} G(x', t) \psi_0(x - x') dx' \\ + \frac{1}{2} \psi_0(x - t) + \frac{1}{2} \psi_0(x + t), \end{aligned} \quad (36)$$

where P refers to the principal value of the integral [38]. To see the proof of Eq. (36), and other details, refer to Appendix D. For the initial condition $\psi_0(x)$, we choose a wave function with compact support in the interval $[-\delta/2, \delta/2]$. The simplest case is to assume that the wave function is initially constant and real within this interval. By normalization, we find that the initial wave function is given by $\psi_0(x) = \delta^{-1/2}$ for $|x| < \delta/2$ and zero elsewhere. However, using this initial condition leads to problems because the wave function is nonzero at $x \pm \delta$, resulting in four singularities in the principal value of Eq. (36) at $|x - t| \pm \delta$. To address these singularities, we need to assume that the initial wave function is zero at $x \pm \delta$, which requires the wave function to be continuous. Our alternative initial condition is a harmonic wave function:

$$\psi_0(x) = \sqrt{\frac{2}{\delta}} \cos \frac{\pi x}{\delta} (|x| < \delta/2). \quad (37)$$

For $m > 0$, we calculate the wave function based on the numerical integration of the propagator, and the results are shown in Fig. 2(c). Despite the singularity of the propagator at $x = \pm t$, the wave functions are continuous and differentiable everywhere. The propagation of the wave function resembles that in the wave equation, where an initial pulse is split into two equal pulses moving left and right. The key difference is that the pulses are complex and involve both propagation and dissipation. The probability density is calculated as $\rho = \psi^* \psi$ and is shown in Fig. 2(d). The continuity equation is verified by calculating the area under ρ for different values of the parameter ϵ used in the calculation of the principal value of the wave function. It is confirmed that for all the time values shown in Fig. 2, the limit of the integral is very close to 1, with an error below 0.004%. This result is consistent with the continuity equation for the Salpeter equation, as demonstrated by Kowalski and Rembielinski [28].

III. BÄUMER HAMILTONIAN

Now we consider the relativistic diffusion equation (RDE), which is obtained by a Wick rotation, or equivalently, by multiplying the Hamiltonian by the imaginary unit i . The corresponding Hamiltonian has been investigated by Bäumer *et al.* [37] and will be referred to here as the Bäumer Hamiltonian,

$$H_B(p) = i\sqrt{m^2 + p^2} - im. \quad (38)$$

Substituting this Hamiltonian into the Schrödinger equation (A1), and using Eq. (A13), the RDE is obtained as the following fractional differential equation:

$$2D \left[\sqrt{1 + \frac{\partial^2}{\partial x^2}} - 1 \right] P = \frac{\partial P}{\partial t}, \quad (39)$$

where $D \equiv \frac{m}{2}$, and P is the probability density in the stochastic space, equivalent to the wave function in the quantum space. The quantum mechanics mathematical formalism allows a simple solution of this fractional differential equation by solving the propagator that in this case is the Green function of the RDE.

The propagator in Eq. (A7) for the Hamiltonian given by Eq. (38) becomes

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\sqrt{m^2 + p^2}t + m t + i p x} dp, \quad (40)$$

whose components are

$$G_{\pm}(x, t) = \frac{1}{2\pi} \int_0^{\infty} e^{-\sqrt{m^2 + p^2}t + m t \pm i p x} dp. \quad (41)$$

We consider different cases to discuss the solutions as shown below.

A. Case $m > 0$ and $t \gg 1$ ²

The classical limit is the same as the asymptotic solution when $t \rightarrow \infty$. In such a case, the propagator (40) becomes

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(p^2/2m)t + ipx} dp. \quad (42)$$

The solution corresponds to a Gaussian distribution

$$G(x, t) = \sqrt{\frac{m}{2\pi t}} e^{-\frac{mx^2}{2t}}. \quad (43)$$

We can also recover the Green function of the diffusion equation by making the transformation $t \rightarrow it$ in the traditional Schrödinger equation and taking $1/(2m) \rightarrow D$:

$$G(x, t) \rightarrow \frac{1}{\sqrt{\pi Dt}} e^{-\frac{x^2}{4Dt}}. \quad (44)$$

B. Integral representation for $m > 0$ and $0 \leq x < \infty$ (inner solution)

For the general case, the components in Eq. (41) can be summed up to obtain

$$G(x, t) = \frac{e^{mt}}{\pi} \int_0^{\infty} e^{-\sqrt{m^2+p^2}t} \cos(px) dp. \quad (45)$$

In the paper of Bäumer *et al.* [37], the integral is solved using the inverse Fourier transformation. However, we note that the integrand has a strong oscillatory behavior, such as $x \rightarrow \infty$. This integral is therefore the *inner solution* of the propagator that can be applied to moderate values of x but will not effectively converge at the tails. An outer solution is sought in the next section using complex variable analysis.

C. Integral representation for $m > 0$ and $x > 0$ (outer solution)

We write the two terms of the propagator as Eq. (15),

$$G_{\pm}(x, t) = \frac{1}{2\pi} \int_0^{\infty} e^{\Phi_{\pm}(p)} dp,$$

and the asymptotic behavior of $\Phi_{\pm}(z) = -iH_B(z)t \pm xz$ for $z = Re^{i\theta}$ as $R \rightarrow \infty$ follows:

$$\lim_{R \rightarrow \infty} \text{Re} \Phi(Re^{i\theta}) = \lim_{R \rightarrow \infty} -R(t \cos \theta \pm x \sin \theta). \quad (46)$$

The condition given by Eq. (46) is satisfied in the first quadrant of the complex plane for G_+ and in the fourth quadrant for G_- . The integration of G_+ over the first quadrant gives

$$\begin{aligned} G_+(x, t) &= \frac{1}{2\pi} \int_0^{\infty} e^{-\sqrt{m^2+p^2}t + ipx} dp \\ &= \frac{e^{mt}}{2\pi} \int_0^{\infty} e^{-\sqrt{m^2+(iq)^2}t + i(iq)x} d(iq) \\ &= \frac{ie^{mt}}{2\pi} \left(\int_0^m e^{-\sqrt{m^2-q^2}t - qx} dq + \int_m^{\infty} e^{-i\sqrt{q^2-m^2}t - qx} dq \right). \end{aligned}$$

The integration of G_- over the four quadrants results in

$$\begin{aligned} G_-(x, t) &= \frac{1}{2\pi} \int_0^{\infty} e^{-\sqrt{m^2+p^2}t + ipx} dp \\ &= \frac{e^{mt}}{2\pi} \int_0^{\infty} e^{-\sqrt{m^2+(-iq)^2}t - i(-iq)x} d(-iq) \\ &= \frac{-ie^{mt}}{2\pi} \left(\int_0^m e^{-\sqrt{m^2-q^2}t - qx} dq + \int_m^{\infty} e^{i\sqrt{q^2-m^2}t - qx} dq \right). \end{aligned}$$

Adding both parts and the full propagator becomes

$$G(x, t) = \frac{-e^{mt}}{\pi} \int_m^{\infty} \sin(\sqrt{q^2 - m^2}t) e^{-qx} dq. \quad (47)$$

Interestingly, using the identity $\sinh(iz) = i \sin(z)$ and replacing t by it , we can recover the outer-cone solution of the Salpeter equation, Eq. (22).

D. Case $m = 0$

For massless stochastic particles, the components of the propagator in Eq. (41) become

$$G_{\pm}(x, t) = \frac{1}{2\pi} \int_0^{\infty} e^{-pt \pm ipx} dp. \quad (48)$$

Both integrals can be analytically integrated. Summing up, the solution is

$$G_{m=0}(x, t) = \frac{1}{\pi} \frac{t}{t^2 + x^2}. \quad (49)$$

This is the Cauchy distribution with the scale parameter given by t .

E. Case $x \gg t$

The behavior of the tail of the Bäumer distribution can be obtained by using $x \gg t$ in Eq. (47),

$$G(x, t) = \frac{-e^{m(t-x)}}{\pi}. \quad (50)$$

Interestingly, the Bäumer distribution exhibits exponential-tailed behavior, which is not the case for either the Cauchy distribution or the Gaussian distribution.

F. Analytical solution

The analytical solution of the Bäumer propagator is obtained by substituting the imaginary Hamiltonian into the Salpeter propagator,

$$\begin{aligned} G(x, t) &= \frac{1}{\pi} \int_0^{\infty} e^{-i^2(\sqrt{m^2+p^2}-m)t} \cos(px) dp \\ &= \frac{1}{\pi} e^{-mt} \int_0^{\infty} e^{\sqrt{m^2+p^2}t} \cos(px) dp. \end{aligned} \quad (51)$$

Using the identity in Eq. (28), we have $\alpha = -t$, $\beta = t$, and $\gamma = x$, and the solution to the integral can be written as

$$G(x, t) = -\frac{mt e^{-mt}}{\pi \sqrt{x^2 + t^2}} K_1(m \sqrt{x^2 + t^2}). \quad (52)$$

Note that the solution of the propagator defines a diffusion process with a transition from Cauchy diffusion to Gaussian

²In dimensional form, the case is $t\hbar c^2/D \gg 1$.

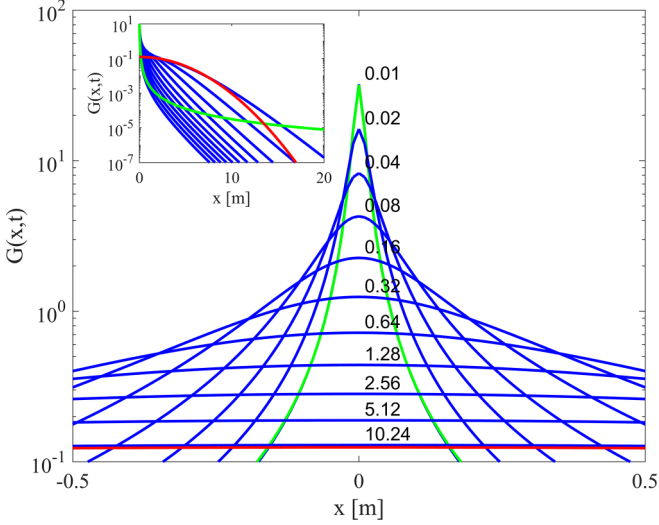


FIG. 4. Solution of the B umer propagation that describes the relativistic diffusion propagator. $G(x, t)$ is plotted in blue against x for different values of t . The propagator is compared to the Cauchy distribution (green) and Gaussian distribution (red). The inset shows the exponential law of the tails.

diffusion, as shown in Fig. 4. The propagator (i.e., the Green function) asymptotically converges to a Cauchy distribution as $t \rightarrow 0$, a Gaussian distribution as $t \rightarrow \infty$, and an exponential function when $x \rightarrow \infty$.

The B umer propagator defines a diffusion process that is more sophisticated than Gaussian diffusion. In most cases, the diffusivity is characterized by either a power-law in the evolution of the peak, $G(0, t) \sim t^{-1/\alpha}$, or the power-law of the evolution of the second moment, $\sqrt{\langle x^2 \rangle} \sim t^{1/\alpha}$. The diffusion process corresponds to normal diffusion when $\alpha = 2$, superdiffusion when $\alpha > 2$, and subdiffusion when $\alpha < 2$. Figure 5 shows the time evolution of both the peak and the second moment of the propagator. The evolution of the peak shows a crossover between subdiffusion ($\alpha = 1$) for

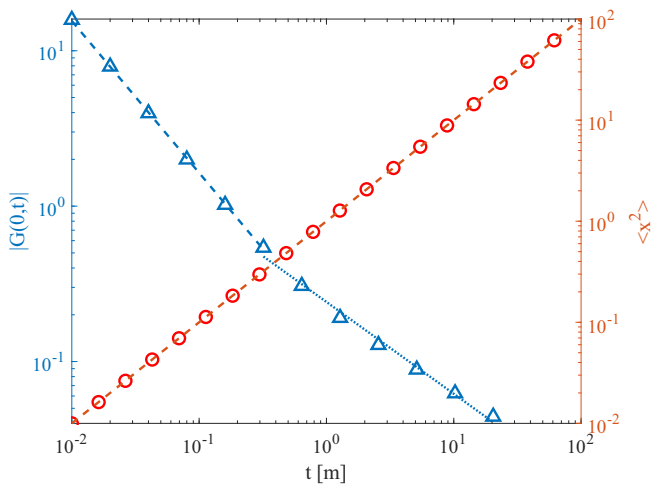


FIG. 5. The time evolution of the peak $G(0, t)$ (blue) and second moment $\langle x^2 \rangle$ (red) of the B umer propagator. The solid blue line corresponds to power law $y = t$.

small times and normal diffusion ($\alpha = 2$) for large times. In contrast, the second moment exhibits normal diffusion at all times. This complexity of the diffusion process, with crossover regions over space-time, is also observed in the temporal evolution of the probability density function (PDF) of the predicted price returns [35]. However, the exponents of the stock market diffusion process differ from those of the B umer diffusion process. In particular, the diffusion process in stock markets is not only fractional, but also nonlinear [40,41]. In addition, the tails of the distribution of stock market returns are power-laws and not exponential as in the B umer distribution; see the inset of Fig. 4. Thus, the application of quantum mechanics to stock market prediction remains limited due to current constraints in mathematically describing the nonlinearity of quantum mechanics. The Wick rotation links stochastic equations and quantum mechanics, offering a path to solve complex stochastic problems (e.g., porous media fractional equation, nonlinear Fokker-Planck equation) and then address corresponding quantum problems. The Ito lemma connects the Fokker-Planck equation (describing PDF evolution) to stochastic processes. Thus, the Wick rotation converts quantum equations to diffusion equations, linked to stochastic equations via Ito's lemma. This Ito-Wick transformation unifies three disciplines, as explored here.

IV. CONCLUSIONS

In this paper, we present the numerical, analytical, and asymptotic solutions for the Salpeter propagator in the context of relativistic quantum mechanics for a free particle with no spin, alongside its stochastic counterpart, the B umer propagator. The solution of the B umer propagator asymptotically converges to a Cauchy distribution at small times and to a Gaussian distribution at large times. This anomalous crossover diffusion is characteristic of stock market fluctuations. The Salpeter propagation is converted into the B umer Green function by either transforming time using imaginary time (Wick rotation) or transforming the real Hamiltonian into a purely imaginary one (complex Hamiltonian). The solution of this quantum-mechanical propagator exhibits several important properties, including nonzero values outside the light cone, propagation with diffusion around the light cone, and a singularity at the light cone. Special handling is required to address this singularity in the calculation of the wave function using the principal value of integrals. An initial wave function with compact support on discontinuities at the edges allows us to remove the singularity, leading to well-posed wave functions. The conservation of probability is verified by ensuring that the integral of the probability density remains equal to 1 at all times.

This paper has focused on applying complex variable analysis to derive solutions, and conducting a careful numerical test of these solutions. By transformation of the integrals in the complex plane, stable numerical solutions are achieved that perfectly match the analytical solutions. The same techniques can be extrapolated to more complex problems where analytical solutions are not available. We addressed the singularity of the Salpeter propagator on the light cone. Yet, well-posed solutions of the wave function are obtained by deforming the integral to avoid the singularities and by choosing appropriate

boundary conditions: The initial wave function with compact support is continuous at its boundaries.

Future work will involve the extension of these solutions to nonlinear equations within the context of nonlinear quantum mechanics, quantum field theories, and nonlinear stochastic dynamics. Exploring the links between these formulations can benefit all these areas that are usually investigated separately. In the same way that the Ito lemma connects stochastic dynamics to the Fokker-Planck equation (FPE) [42], the Wick rotation establishes a link between the FPE and quantum mechanics.

APPENDIX A: QUANTUM PROPAGATOR

We can write the Schrödinger equation for a particle in the bracket notation:

$$\hat{H}|\psi\rangle = i\hbar\partial_t|\psi\rangle; \quad |\psi(t_0)\rangle = |\psi_0\rangle. \quad (\text{A1})$$

Here, the $\hat{H}$ is the Hamiltonian of the particle, $|\psi\rangle$ is the state of the particle at time t , and $|\psi_0\rangle$ is the initial condition at time t_0 . Let us define the temporal evolution operator by

$$\hat{U}(t_0, t)|\psi(t_0)\rangle = |\psi(t)\rangle. \quad (\text{A2})$$

Using Eq. (A1), the evolution equation of $\hat{U}$ can be written as

$$i\hbar\partial_t\hat{U} = \hat{H}\hat{U}; \quad \hat{U}(t_0) = \hat{1}. \quad (\text{A3})$$

Since the particle is in a state $|\psi(t_0)\rangle$ at $t = t_0$, the operation of temporal evolution brings the particle to a new state $|\psi(t)\rangle$. Then, the probability of finding the particle in a position x is calculated by a bracket operation $\psi(t, x) = \langle x|\psi(t)\rangle$. This is the wave function of the particle, which can be calculated by multiplying Eq. (A2) by $\langle x|$,

$$\psi(t, x) = \langle x|\hat{U}(t_0, t)|\psi(t_0)\rangle. \quad (\text{A4})$$

Let us introduce the completeness operator for the observable $\hat{x}$,

$$\hat{1} = \int dx_0 |x_0\rangle\langle x_0|. \quad (\text{A5})$$

Inserting this operator into Eq. (A4), we obtain

$$\begin{aligned} \psi(t, x) &= \langle x|\hat{U}(t_0, t)\hat{1}|\psi(t_0)\rangle \\ &= \int dx_0 \langle x|\hat{U}(t_0, t)|x_0\rangle\langle x_0|\psi(t_0)\rangle \\ &= \int dx_0 G(x, t, x_0, t_0)\psi(t_0, x_0). \end{aligned} \quad (\text{A6})$$

Here, we define the *propagator* by the function

$$G(x, t, x_0, t_0) = \langle x|\hat{U}(t_0, t)|x_0\rangle. \quad (\text{A7})$$

According to Eq. (A17), it is noted that the propagation behaves similarly to the Green functions of the partial differential equation. This function can be calculated for some specific cases. If the Hamiltonian is independent of time, the evolution operator of Eq. (A3) can be easily solved to obtain

$$\hat{U}(t, t_0) = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}. \quad (\text{A8})$$

Equation (A8) can be replaced with Eq. (A7) to obtain the explicit expression of the propagator. This requires the introduction of monochromatic wave functions derived from the

eigenstates of the momentum observable $\hat{p}$,

$$|p\rangle = \frac{1}{\sqrt{2\pi}} \int dx_0 e^{\frac{i}{\hbar}px_0}|x_0\rangle. \quad (\text{A9})$$

The wave functions of the eigenstates $|x\rangle$ of the observables $\hat{x}$ are given by Dirac's delta function,

$$\psi_{x_0}(x) = \langle x|x_0\rangle = \delta(x - x_0). \quad (\text{A10})$$

In agreement with Eq. (A5), the property of the Dirac function holds for any function $f(x)$,

$$\int dx f(x)\delta(x - x_0) = f(x_0). \quad (\text{A11})$$

The corresponding wave function of the eigenstates $|p_0\rangle$ of the observable $\hat{p}$ is given by

$$\psi_{p_0}(x) = \langle x|p_0\rangle = \frac{1}{\sqrt{2\pi}} \int dx_0 e^{\frac{i}{\hbar}px_0} \langle x|x_0\rangle = \frac{1}{\sqrt{2\pi}} e^{\frac{i}{\hbar}p_0x}. \quad (\text{A12})$$

Let us note that $i\hbar\partial_x\psi_{p_0}(x) = p\psi_{p_0}(x)$, which indicates that

$$\langle x|\hat{p}|p_0\rangle = -i\hbar\partial_x\langle x|k\rangle. \quad (\text{A13})$$

The completeness relation for the $\hat{p}$ observable also holds,

$$\hat{1} = \int dp_0 |p_0\rangle\langle p_0|. \quad (\text{A14})$$

Let us assume a Hamiltonian of the free particle

$$\hat{H} = \hat{H}(p). \quad (\text{A15})$$

Using Eq. (A14) in the propagator [Eq. (A7)] gives

$$\begin{aligned} G(x, t, x_0, t_0) &= \langle x|e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}|x_0\rangle \\ &= \int dp \langle x|e^{-\frac{i}{\hbar}\hat{H}(p)(t-t_0)}|p\rangle\langle p|x_0\rangle \\ &= \int dp e^{-\frac{i}{\hbar}H(p)(t-t_0)} \langle x|p\rangle\langle p|x_0\rangle \\ &= \frac{1}{2\pi} \int dp \exp\left[-\frac{i}{\hbar}H(p)(t-t_0) + \frac{i}{\hbar}p(x-x_0)\right]. \end{aligned} \quad (\text{A16})$$

Hence the propagator satisfies

$$G(x, t, x_0, t_0) = G(x - x_0, t - t_0).$$

Using this property, and assuming that $t_0 = 0$, we can simplify the calculation of the wave function by Eq. (A6) to

$$\psi(t, x) = \int_{-\infty}^{\infty} G(x - x_0, t)\psi_0(x_0)dx_0, \quad (\text{A17})$$

where $\psi_0(x)$ is the initial condition and

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}H(p)t + i\hbar px} dp. \quad (\text{A18})$$

APPENDIX B: SCALING RELATIONS

In this appendix, we focus on the scaling properties of the Salpeter equation, which helps us to recognize the general features of the model. Note that the full functionality of the Green function is $G(x, t; x', t'; m)$, but without loss of generality we set $x' = t' \equiv 0$ and $G(x, t, m) \equiv G(x, t; 0, 0; m)$,

keeping in mind that eventually we should replace $x \rightarrow x - x'$ and $t \rightarrow t - t'$. We have the following integral expression:

$$G(x, t, m) \equiv \frac{1}{2\pi} e^{imt} \int_0^\infty e^{-i\sqrt{m^2+p^2}t} \cos(px) dp. \quad (\text{B1})$$

One can easily show by inspection that for any positive scaling parameter $\lambda > 0$, we have

$$G(\lambda x, \lambda t, m) = \lambda^{-1} G(x, t, \lambda m), \quad (\text{B2})$$

so that under the transformation

$$x \rightarrow \lambda x, \quad t \rightarrow \lambda t, \quad m \rightarrow \lambda^{-1} m \quad (\text{B3})$$

we have $G \rightarrow \lambda^{-1} G$. Therefore, we expect the following identity:

$$G(x, t) = m \tilde{G}[m(ax + bt)], \quad (\text{B4})$$

where $\tilde{G}$ is a general function that should be determined using other methods, and a and b are some constant complex numbers. The argument of $\tilde{G}$ guarantees the invariance of $\tilde{G}$ under scaling transformation. The limit $m \rightarrow 0$ helps to identify this function. To this end, we add a vanishingly small quantity to time, i.e., $t \rightarrow t - i0^+$ (which guarantees that the integral is well-defined in the UV limit $p \rightarrow \infty$), so that

$$G(x, t, m=0) = \frac{1}{2\pi} \left[\frac{i}{x-t+i0^+} - \frac{i}{x+t-i0^+} \right]. \quad (\text{B5})$$

Using the identity

$$\frac{1}{X+i0^\pm} = P \frac{1}{X} \mp i\pi \delta(X), \quad (\text{B6})$$

where P stands for principal value, we obtain

$$G(x, t, m=0) = \frac{it}{\pi(x^2 - t^2)} + \frac{1}{2} \delta(x-t) + \frac{1}{2} \delta(x+t), \quad (\text{B7})$$

where $\delta(x)$ is a Dirac delta function, and the first term on the right is P . One may compare this result with the orthogonality identity for plane waves,

$$\frac{1}{2\pi} \int_{-\infty}^\infty e^{ip(x \pm t)} dp = \frac{1}{\pi} \lim_{p \rightarrow \infty} \frac{\sin p(x \pm t)}{x \pm t} = \delta(x \pm t), \quad (\text{B8})$$

from which we realize that the first term on the right-hand side of Eq. (B7) is due to the presence of *absolute value* in the dispersion relation $E = |p|$ in the $m \rightarrow 0$ limit. From this limit, we realize that $a = 1$ and $b = \pm 1$. We additionally find that for very small m , we have

$$\tilde{G}(x, t, m) = \frac{1}{2\pi} \left[\frac{i}{m(x-t)+i0^+} - \frac{i}{m(x+t)-i0^+} \right]. \quad (\text{B9})$$

The other quantity of interest is $\langle x^2 \rangle$. From the symmetry Eq. (B2) one finds that

$$\langle x^2 \rangle = m^{-2} \tilde{f}(mt) = t^2 \tilde{g}(mt), \quad (\text{B10})$$

where $\tilde{f}(y) = y^2 \tilde{g}(y)$ are general smooth functions. For the massless free-particle limit, we know

$$\langle x^2 \rangle_{\text{massless}} = t^2, \quad (\text{B11})$$

from which we see

$$\tilde{g}(y)|_{m \rightarrow 0} \rightarrow 1. \quad (\text{B12})$$

APPENDIX C: INTEGRAL TRANSFORMATION THEOREM

Analytical integration of improper functions is available only in simple cases. If the integrand poses strong oscillatory behavior, numerical integration is also inconvenient and often lacks accuracy. Using the Cauchy theorem of complex variable analysis, it is possible to transform integrals with oscillatory behavior to integrands with smooth exponential decreases that are easier to integrate.

Theorem C1. Suppose that $f(z) = e^{\Phi(z)}$ is an analytical function in the first quarter of the complex plane:

$$D_1 = \{z = \rho e^{i\theta} = p + iq; 0 \leq \theta \leq \pi/2\}$$

and $\Phi(z)$, with $z = \rho e^{i\theta} = p + iq$, satisfies

$$\lim_{\rho \rightarrow \infty} \text{Re} \Phi(\rho e^{i\theta}) = -\infty \quad z \in D_1. \quad (\text{C1})$$

The integral of $f(z)$ over the semireal axis $\mathbb{R} \cap D_n$ can be transformed into an integral over the semi-imaginary axis $\mathbb{I} \cap D_1$ as

$$\int_0^\infty e^{\Phi(p)} dp = \int_0^\infty e^{\Phi(iq)} i dq. \quad (\text{C2})$$

Proof. Since $f(z)$ is analytical in the first quadrant, its integral over any of the circular sectors on the first quadrant is zero,

$$\oint_{D_1} f(z) dz = \oint_{D_1} e^{\Phi(z)} dz = 0. \quad (\text{C3})$$

Applying this property to a circular sector of infinite radius,

$$\int_0^\infty e^{\Phi(p)} dp + \lim_{\rho \rightarrow \infty} \int_0^{\pi/2} e^{\Phi(\rho e^{i\theta})} d\theta + \int_\infty^0 e^{\Phi(iq)} i dq = 0.$$

Thus,

$$\int_0^\infty e^{\Phi(p)} dp = \int_0^\infty e^{\Phi(iq)} i dq - \lim_{\rho \rightarrow \infty} \int_0^{\pi/2} e^{\Phi(\rho e^{i\theta})} d\theta.$$

The second term is expected to vanish under the condition Eq. (C2). ■

Now, we provide a similar theorem for the four quadrants of the complex plane.

Theorem C2. If $f(z) = e^{\Phi(z)}$ is an analytical function in the fourth quarter of the complex plane,

$$D_4 = \{z = \rho e^{i\theta} = p + iq; -\pi/2 \leq \theta \leq 0\}$$

and $\Phi(z)$, with $z = \rho e^{i\theta} = p + iq$ satisfies

$$\lim_{\rho \rightarrow \infty} \text{Re} \Phi(\rho e^{i\theta}) = -\infty \quad z \in D_4. \quad (\text{C4})$$

The integral of $f(z)$ over the semireal axis $\mathbb{R} \cap D_4$ can be transformed into an integral over the semi-imaginary axis $\mathbb{I} \cap D_4$ as

$$\int_0^\infty e^{\Phi(p)} dp = \int_0^\infty e^{\Phi(-iq)} (-i) dq. \quad (\text{C5})$$

Proof. The theorem will be proof for the fourth quadrant. Since $f(z)$ is analytical in the four quadrants, its integral over any of the circular sectors on the first quadrant is zero,

$$\oint_{D_4} f(z)dz = \oint_{D_4} e^{\Phi(z)} dz = 0.$$

Applying this property to a circular sector of infinite radius,

$$\int_0^\infty e^{\Phi(p)} dp + \lim_{\rho \rightarrow \infty} \int_0^{-\frac{\pi}{2}} e^{\Phi(\rho e^{i\theta})} d\theta + \int_{-\infty}^0 e^{\Phi(iq)} idq = 0.$$

Thus, replacing q by $-q$ in the last integral,

$$\int_0^\infty e^{\Phi(p)} dp = \int_0^\infty e^{\Phi(-iq)} (-idq) - \lim_{\rho \rightarrow \infty} \int_0^{-\frac{\pi}{2}} e^{\Phi(\rho e^{i\theta})} d\theta.$$

The second term is expected to vanish under the condition (C4). ■

APPENDIX D: THEOREM OF RESIDUES

In this appendix, we provide a proof of Eq. (36). The main focus is to calculate the following integral:

$$\psi^*(t, x) = \int_C G(z, t) \psi_0(x - z) dz, \quad (D1)$$

where C is the contour in the complex plane shown in Fig. 3. The function $G(z, t)$ is the analytical continuation of the propagator [Eq. (11)] in the complex plane:

$$G(z, t) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-iH_S(p)t + ipz} dp, \quad (D2)$$

and the function $\psi_0(z)$ is the initial wave function [Eq. (37)]

$$\psi_0(z) = \sqrt{\frac{2}{\delta}} \cos \frac{\pi z}{\delta} (|z| < \delta/2). \quad (D3)$$

The integral (36) diverges because of the nonintegrable poles on the integration contour. Since these poles are of the first order, the Cauchy principal value (P) of the integral exists and corresponds to avoiding the poles with semicircular arcs. To prove this, we use the Cauchy residue theorem, which states that if the integrand is analytic on a simply connected domain except for a finite number of isolated singularities, then the integral of that around a closed curve is equal to $2\pi i$ times the sum of the residues of the integrand at the singular points inside the closed curve. First, we consider the case $m = 0$ and show that the P is

$$\text{P} \int_{-\infty}^\infty G(y, t) \psi_0(x - y) dy = \frac{1}{2} [\psi_0(x - t) + \psi_0(x + t)]. \quad (D4)$$

The contour in Fig. 3 is chosen so that there are two poles inside it and the little circles around each of the poles are so small. The contour is

$$C = C^R + C_1^\epsilon + C_2^\epsilon + C_{-\infty}^\infty, \quad (D5)$$

where $C_{-\infty}^\infty$ is the integral in Eq. (D4), C_i^ϵ , $i = 1, 2$, are the contributions due to integrating over the poles at $\pm t$, and C^R is the (vanishing) contribution due to the large arc.

To integrate around the pole $-t$, let us parametrize $z = -t + \epsilon e^{i\theta}$, and then $dz = i\epsilon e^{i\theta} d\theta$. For the case $m = 0$,

by considering Eq. (26), integrating around the associated semicircle yields

$$\begin{aligned} & \int_{C_1^\epsilon} G(z, t) \psi_0(x - z) dz \\ &= \lim_{\epsilon \rightarrow 0} \int_{\pi}^{2\pi} G(-t + \epsilon e^{i\theta}, t) \psi_0(x + t - \epsilon e^{i\theta}) i\epsilon e^{i\theta} d\theta \\ &= \lim_{\epsilon \rightarrow 0} \frac{-t}{\pi} \int_{\pi}^{2\pi} \frac{\psi_0(x + t - \epsilon e^{i\theta})}{\epsilon e^{i\theta} (-2t + \epsilon e^{i\theta})} \epsilon e^{i\theta} d\theta \\ &= \frac{1}{2} \psi_0(x + t). \end{aligned} \quad (D6)$$

Similarly, to integrate around the pole t , we parametrize $z = t + \epsilon e^{i\theta}$, and then $dz = i\epsilon e^{i\theta} d\theta$. By considering Eq. (26), integrating around the other semicircle yields

$$\int_{C_2^\epsilon} G(z, t) \psi_0(x - z) dz = \frac{1}{2} \psi_0(x - t). \quad (D7)$$

Also, the integration along the semicircular portion C^R of the contour C vanishes as the radius of integration becomes very large, i.e., as $R \rightarrow \infty$,

$$\int_{C^R} G(z, t) \psi_0(x - z) dz = 0, \quad (D8)$$

and by considering the counter (D5), this completes the proof of Eq. (36).

To prove Eq. (D4), since there are two poles inside of the contour, we have by Cauchy's residue theorem,

$$\int_C G(z, t) \psi_0(x - z) dz = 2i\pi [\text{Res}(-t) + \text{Res}(t)]. \quad (D9)$$

By using Theorem IV (p. 557) from [38] and Eqs. (D6) and (D7),

$$\text{Res}(\pm t) = \frac{1}{2i\pi} \psi_0(x \mp t), \quad (D10)$$

and by considering the counter (D5), this completes the proof of Eq. (D4).

Now, we consider the case in which $m \neq 0$. We follow a similar scenario on the same contour. Therefore, under these conditions, Eqs. (D5), (D8), and (D9) work for our case, and we should investigate Eq. (D4). To achieve this, we applied the analytical solution of the Salpeter propagator, as described in Eq. (33), for integrals (D6) and (D7). We emphasize the case $m \neq 0$ by denoting

$$G^m(z, t) = \begin{cases} \frac{imt e^{imt}}{\pi \sqrt{z^2 - t^2}} K_1(m \sqrt{z^2 - t^2}), & |z| > t, \\ -\frac{mt e^{imt}}{2\sqrt{t^2 - z^2}} H_1^{(1)}(-m \sqrt{t^2 - z^2}), & |z| < t, \end{cases} \quad (D11)$$

where $H_1^{(1)}$ is the Hankel function of the first kind, and K_1 is the modified Bessel function of the second kind.

Integrating around the first semicircle yields

$$\begin{aligned} & \int_{C_1^+} G^m(z, t) \psi_0(x - z) dz \\ &= \lim_{\epsilon \rightarrow 0} \int_{\pi}^{2\pi} G^m(-t + \epsilon e^{i\theta}, t) \psi_0(x + t - \epsilon e^{i\theta}) i \epsilon e^{i\theta} d\theta \\ &= \lim_{\epsilon \rightarrow 0} \frac{-imt e^{imt}}{\pi} \int_{\pi}^{2\pi} \frac{K_1(m\epsilon^{1/2} e^{i\theta/2} \sqrt{-2t + \epsilon e^{i\theta}})}{\epsilon^{1/2} e^{i\theta/2} \sqrt{-2t + \epsilon e^{i\theta}}} \\ & \quad \times \psi_0(x + t - \epsilon e^{i\theta}) i \epsilon e^{i\theta} d\theta. \end{aligned}$$

Applying the fact that $K_1(y) \rightarrow \frac{1}{y}$ as $y \rightarrow 0$, we get

$$\int_{C_1^+} G^m(z, t) \psi_0(x - z) dz = \frac{1}{2} \psi_0(x + t) e^{imt}. \quad (D12)$$

Similarly, integrating around the second semicircle yields

$$\int_{C_2^+} G^m(z, t) \psi_0(x - z) dz = \frac{1}{2} \psi_0(x - t) e^{imt}. \quad (D13)$$

We can easily acquire

$$\text{Res}(\pm t) = \frac{e^{imt}}{2i\pi} \psi_0(x \mp t), \quad (D14)$$

and then the P is

$$\begin{aligned} P \int_{-\infty}^{\infty} G(y, t) \psi_0(x - y) dy \\ = \frac{1}{2} (\psi_0(x - t) + \psi_0(x + t)) e^{imt}. \end{aligned} \quad (D15)$$

APPENDIX E: ANALYTICAL EXPRESSIONS BASED ON PERTURBATIVE EXPANSION

In this appendix, we present an analytical expansion of the Salpeter integral, which serves as a perturbative expansion. It helps to realize the analytical form of the function around, e.g., the light cone and other regions of interest, and it also shows the internal structure and consistency of the analytical expressions. We follow the expansion scheme presented according to an *ad hoc* weight function. In this method, we multiply the integrand by a weight function $f(p, p_0)$, which makes the approximate integration feasible in an appropriate limit, and p_0 is a characteristic cutoff momentum dividing the integral domain for two intervals, each being treated differently in the perturbation expansion. In our case, $p_0 \equiv m$ is the characteristic momentum that separates “large” and “short” p values. The choice of $f(p, p_0)$ is not unique and can be any form depending on the form of the integral; for example, it can be either ($p_0 \equiv m$)

$$f(p, p_0) = \exp \left[-\left(\frac{p}{p_0} \right)^2 \right]$$

or

$$f(p, p_0) = \Theta(p - p_0), \quad (E1)$$

where $\Theta(x)$ is a step function. In this paper, choose the second “cut” function, and $p_0 \equiv m$. We then cast the formula Eq. (2)

to the following:

$$\begin{aligned} G(x, t, 0, 0) &= \int_0^{\infty} dp [f(p, p_0) + 1 - f(p, p_0)] \\ & \quad \times \exp[-itH(p)] \cos(px) \\ &\equiv G_1(x, t) + G_2(x, t). \end{aligned} \quad (E2)$$

The integrand of function

$$G_1(x, t) = \int_0^{\infty} dp f(p, p_0) \exp[-itH(p)] \cos(px) \quad (E3)$$

can be expanded for small p values ($p \ll p_0$), while

$$G_2(x, t) = \int_0^{\infty} dp [1 - f(p, p_0)] \exp[-itH(p)] \cos(px) \quad (E4)$$

should be expanded around small $\frac{1}{p} \ll \frac{1}{p_0}$. We end with ($\epsilon_1 \equiv \frac{p}{m}$, and $g_1(mt; \epsilon_1) \equiv \exp[-imt(\sqrt{1 + \epsilon_1^2} - 1)$, $\rho \equiv mx$)

$$\begin{aligned} G_1(x, t) &= m \int_0^1 d\epsilon_1 e^{-imt} g_1(mt; \epsilon_1) \cos(\epsilon_1 mx) \\ &= m \sum_{n=0}^{\infty} e^{-imt} \frac{g_1^{(n)}(mt)}{n!} \int_0^1 d\epsilon_1 \epsilon_1^n \cos(\epsilon_1 mx) \\ &= m e^{-imt} \sum_{n=0}^{\infty} \frac{g_1^{(n)}(mt)}{n!} \text{Re} \left(-i \frac{\partial}{\partial \rho} \right)^n \int_0^1 d\epsilon_1 e^{i\rho \epsilon_1} \\ &= m e^{-imt} \sum_{n=0}^{\infty} \frac{g_1^{(n)}(mt)}{n!} f_n(mx), \end{aligned} \quad (E5)$$

where

$$g_1^{(n)}(mt) \equiv \frac{\partial^n}{\partial \epsilon_1^n} g_1(mt; \epsilon_1) |_{\epsilon_1=0}, \quad (E6)$$

and

$$\begin{aligned} f_n(\rho) &\equiv \text{Re} \left[\left(-i \frac{\partial}{\partial \rho} \right)^n \left(\frac{e^{i\rho} - 1}{i\rho} \right) \right] \\ &= \begin{cases} (-1)^{\frac{n+1}{2}} \left(\frac{\partial}{\partial \rho} \right)^n \left(\frac{\cos \rho - 1}{\rho} \right), & n = \text{odd}, \\ (-1)^{\frac{n}{2}} \left(\frac{\partial}{\partial \rho} \right)^n \left(\frac{\sin \rho}{\rho} \right), & n = \text{even}. \end{cases} \end{aligned} \quad (E7)$$

Note that $g_1^{(n)} = 0$ for all *odd* n values, while for the even ones we have

$$\begin{aligned} g_1^{(0)}(y) &= 1, \\ g_1^{(2)}(y) &= -iy, \\ g_1^{(4)}(y) &= 3y(i - y), \\ g_1^{(6)}(y) &= 15y(-3i + 3y + iy^2), \\ g_1^{(8)}(y) &= 15y(15i + 15y - 6iy^2 + y^3). \\ &\dots \end{aligned} \quad (E8)$$

The asymptotic behavior of $g_1^{(2n)}$ is

$$g_1^{(2n)}(y) \rightarrow \begin{cases} A_n^{(1)} y^n, & y \gg 1, \\ B_n^{(1)} y, & y \ll 1, \end{cases} \quad (E9)$$

where $A_n^{(1)}$ and $B_n^{(1)}$ are two expansion coefficients that are given in Eq. (E8).

$f_n(\rho)$ can also be written in terms of the generalized hypergeometric function as follows:

$$f_n(\rho) = \frac{{}_1F_2\left[\frac{n+1}{2}, \left\{\frac{1}{2}, \frac{n+3}{2}\right\}, -\frac{\rho^2}{4}\right]}{n+1}. \quad (\text{E10})$$

The explicit form of f_n is

$$\begin{aligned} f_0(\rho) &= \frac{\sin \rho}{\rho}, \\ f_1(\rho) &= \frac{\cos \rho - 1 + \rho \sin \rho}{\rho^2}, \\ f_2(\rho) &= \frac{2\rho \cos \rho + (\rho^2 - 2) \sin \rho}{\rho^3}, \\ f_3(\rho) &= \frac{6 + 3(\rho^2 - 2) \cos \rho + \rho(\rho^2 - 6) \sin \rho}{\rho^4}, \\ f_4(\rho) &= \frac{4\rho(\rho^2 - 6) \cos \rho + (24 - 12\rho^2 + \rho^4) \sin \rho}{\rho^5}, \\ &\dots \end{aligned} \quad (\text{E11})$$

Note that in the limit $\rho \rightarrow 0$ we have

$$\begin{aligned} f_n(\rho) &= \text{Re} \left[\left(-i \frac{\partial}{\partial \rho} \right)^n \sum_{m=1}^{\infty} \frac{1}{m!} (i\rho)^{m-1} \right] \\ &= \text{Re} \left[\frac{n!}{(n+1)!} + i \frac{(n+1)!}{(n+2)!} \rho \right. \\ &\quad \left. + \frac{(n+2) \cdots 4 \times 3}{(n+3)!} \rho^2 + O(\rho^3) \right] \\ &= \frac{1}{n+1} - \frac{\rho^2}{2(n+3)} + O(\rho^3). \end{aligned} \quad (\text{E12})$$

For the large ρ limit, one can see by inspection that

$$f_n(\rho)|_{\rho \gg 1} \rightarrow a_n \frac{\sin \rho}{\rho}, \quad (\text{E13})$$

where $a_n = 1$ for all n values.

For G_2 we have

$$\begin{aligned} G_2(x, t) &= \int_m^\infty dp e^{-ipt} \sqrt{1 + \left(\frac{p}{\rho}\right)^2} \cos px \\ &= \int_m^\infty dp g_2(mt, \epsilon_2) e^{-ipt} \cos px \\ &= m \sum_{n=0}^{\infty} \frac{g_2^{(n)}(mt)}{n!} \int_1^\infty d\tilde{p} \tilde{p}^{-n} e^{-i\tilde{p}mt} \cos(\tilde{p}mx), \end{aligned} \quad (\text{E14})$$

where $\tilde{p} \equiv p/m$, $\epsilon_2 \equiv \frac{1}{\epsilon_1} = \frac{m}{p}$, and

$$\begin{aligned} g_2(mt, \epsilon_2) &\equiv \exp \left[-i \frac{mt}{\epsilon_2} (\sqrt{1 + \epsilon_2^2} - 1) \right] \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} g_2^{(n)}(mt) \epsilon_2^n = \sum_{n=0}^{\infty} \frac{1}{n!} g_2^{(n)}(mt) \tilde{p}^{-n}, \end{aligned} \quad (\text{E15})$$

and

$$g_2^{(n)}(mt) \equiv \left. \frac{\partial^n}{\partial \epsilon_2^n} g_2(mt, \epsilon_2) \right|_{\epsilon_2=0} \quad (\text{E16})$$

are the coefficients of the Taylor expansion. Notice that when $\epsilon_2 \rightarrow 0$, $\epsilon_2^{-1}(\sqrt{1 + \epsilon_2^2} - 1) \rightarrow 0$.

The coefficients are easily obtained by inspection as follows:

$$\begin{aligned} g_2^{(0)}(y) &= (-i)^0, \\ g_2^{(1)}(y) &= (-i)^1 \frac{y}{2}, \\ g_2^{(2)}(y) &= (-i)^2 \frac{y^2}{2^2}, \\ g_2^{(3)}(y) &= (-i)^3 \frac{y}{2^3} (3! + y^2), \\ g_2^{(4)}(y) &= (-i)^4 \frac{y^2}{2^4} (4! + y^2), \\ g_2^{(5)}(y) &= (-i)^5 \frac{y}{2^5} \left(2 \times 5! + \frac{1}{2} 5! y^2 + y^4 \right), \\ g_2^{(6)}(y) &= (-i)^6 \frac{y^2}{2^6} \left(\frac{5}{2} \times 6! + 5! y^2 + y^4 \right), \\ g_2^{(7)}(y) &= (-i)^7 \frac{y}{2^7} \left(5 \times 7! + \frac{3}{2} 7! y^2 + \frac{7}{4} 5! y^4 + y^6 \right), \\ g_2^{(8)}(y) &= (-i)^8 \frac{y^2}{2^8} \left(7 \times 8! + \frac{14}{3} 7! y^2 + \frac{7}{15} 6! y^4 + y^6 \right), \dots \end{aligned} \quad (\text{E17})$$

Note that

$$g_2^{(n)} \rightarrow \begin{cases} (-i)^n \frac{y^n}{2^n}, & y \gg 1, \\ (-i)^n \frac{n!}{2^n} (A_n^{(2)}) y^{m_n}, & y \ll 1, \end{cases} \quad (\text{E18})$$

where $m_n = 2$ and 1 for n even and odd, respectively, and $A_n^{(2)}$ is an expansion coefficient given in Eq. (E17). To process Eq. (E14), we use the fact that

$$\begin{aligned} I_n(mt, mx) &\equiv \int_1^\infty d\tilde{p} \tilde{p}^{-n} e^{-i\tilde{p}mt} \cos(\tilde{p}mx) \\ &= \frac{1}{2} \text{Ei}_n[im(t-x)] + \frac{1}{2} \text{Ei}_n[im(t+x)], \end{aligned} \quad (\text{E19})$$

where $\text{Ei}_n(y) \equiv \int_1^\infty z^{-n} e^{-yz} dz$ is the exponential integral function. Therefore, we find that

$$\begin{aligned} G_2(x, t) &= \frac{m}{2} \sum_{n=0}^{\infty} \frac{g_2^{(n)}(mt)}{n!} (\text{Ei}_n[im(t-x)] \\ &\quad + \text{Ei}_n[im(t+x)]). \end{aligned} \quad (\text{E20})$$

Note that the singularities of the propagator are due to the presence of the exponential integral functions in $G_2(x, t)$.

Given that

$$\begin{aligned} \text{Ei}_0(z) &= \frac{e^{-z}}{z}, \\ \lim_{z \rightarrow 0} \text{Ei}_1(z) &= -\gamma - \ln z, \\ \lim_{z \rightarrow 0} \text{Ei}_n(z) &= \frac{1}{n-1}, \quad n > 1, \end{aligned} \quad (\text{E21})$$

where γ is an Euler-Mascheroni constant, we find that in the limit $x \rightarrow t$, G_2 becomes singular according to (the same happens for $x = -t$)

$$G_2(x, t) \rightarrow \frac{1}{2i(t-x)} + \frac{im^2 t}{4} \ln[im(t-x)]. \quad (\text{E22})$$

Finally, we find

$$\begin{aligned} G(x, t; 0, 0) &= m \sum_{n=0}^{\infty} \frac{g_1^{(n)}(mt)}{n!(n+1)} {}_1F_2\left[\frac{n+1}{2}, \left\{\frac{1}{2}, \frac{n+3}{2}\right\}, \right. \\ &\quad \left. -\frac{m^2 x^2}{4}\right] + \frac{m}{2} \sum_{n=0}^{\infty} \frac{g_2^{(n)}(mt)}{n!} (\text{Ei}_n[im(t-x)] \\ &\quad + \text{Ei}_n[im(t+x)]). \end{aligned} \quad (\text{E23})$$

This expansion is singular at the Dirac cone, i.e., $x = t$ and $-t$ due to the presence of the exponential integral functions. More precisely, from Eq. (E22) we have

$$\begin{aligned} G(x, t; 0, 0)|_{x \rightarrow \pm t} &= \frac{1}{2i(t \mp x)} + \frac{im^2 t}{4} \ln[im(t \mp x)] \\ &= \frac{1}{2z} + \frac{m^2 z}{8} \ln(mz), \end{aligned} \quad (\text{E24})$$

where we have defined $z \equiv \tau + ix$, $\bar{z} \equiv \tau - ix$, and $\tau \equiv it$ is the Wick-rotated (imaginary) time. In the above equation, we used $z = \tau + ix \approx 2it$ in the limit $x \rightarrow t$ (the same holds for $x \rightarrow -t$ with z replaced by $\bar{z}$). All in all, we find that

$$\begin{aligned} G(x, t; 0, 0)|_{x \rightarrow t} + G(x, t; 0, 0)|_{x \rightarrow -t} \\ = G(x, t, m = 0) + \frac{m^2}{8} [z \ln(mz) + \bar{z} \ln(m\bar{z})]. \end{aligned} \quad (\text{E25})$$

In Appendix B, we discuss the scaling properties of the propagator. Especially, we show that the integral is proportional to the modified Bessel functions of the second type. More precisely,

$$\begin{aligned} G(x, t) &= \frac{imt e^{imt}}{\pi \sqrt{x^2 - t^2}} K_1(m\sqrt{x^2 - t^2}), \quad x^2 \geq t^2 \\ &= -\frac{mt e^{imt}}{2\sqrt{t^2 - x^2}} H_1^{(1)}(-m\sqrt{t^2 - x^2}), \quad x^2 < t^2, \end{aligned} \quad (\text{E26})$$

where $K_1(y)$ is the modified Bessel function of the second type, and $H_1^{(1)}$ is a Hankel function of the first type. This can also be written in terms of the imaginary time τ as follows:

$$\begin{aligned} G(x, \tau) &= \frac{m}{\pi} \cos \theta e^{mr \cos \theta} K_1(mr), \quad x^2 \geq t^2 \\ &= \frac{im}{2} \cos \theta e^{mr' \cos \theta} H_1^{(1)}(mr'), \quad x^2 < t^2, \end{aligned} \quad (\text{E27})$$

where $r \equiv \sqrt{x^2 + \tau^2}$, $r' \equiv \sqrt{t^2 - x^2} = ir$, and $\theta \equiv \tan^{-1}(\frac{x}{\tau})$. For the stochastic process (SP) applications, one may use the first branch, as follows:

$$G_{\text{SP}}(r, \theta) = \frac{m}{\pi} \cos \theta e^{mr \cos \theta} K_1(mr). \quad (\text{E28})$$

This function has been shown in Fig. 6.

APPENDIX F: DIFFERENTIAL EQUATION

Here, we present the connection of the Salpeter equation to the Klein-Gordon differential equation. By differentiating Eq. (11) with respect to x and t , one easily finds ($x, t \neq 0$)

$$(-\partial_x^2 + \partial_t^2 + m^2)G(x, t, m) = 0, \quad (\text{F1})$$

which can be understood using the fact that $H \equiv i\partial_t$ and $p \equiv -i\partial_x$. Above, we considered a complex time $t + i0^+$ to control the integral limits. In an extreme situation, one may Wick-rotate the time, so that $\tau = it$ describes a new *coordination*, which is treated like x as follows ($x, \tau \neq 0$):

$$(\nabla^2 - m^2)G^{(\tau)} = 0, \quad (\text{F2})$$

which is a homogeneous Helmholtz equation in two dimensions for $x, \tau \neq 0$, and $G^{(\tau)}(x, \tau, m) \equiv G(x, -i\tau, m)$, and $\nabla^2 \equiv \partial_x^2 + \partial_\tau^2$. Note that this equation respects the scaling relation given in Eq. (B2). The resulting integral after the Wick rotation describes also the relativistic probability density of stochastic processes. In terms of this quantity, we have

$$G^{(\tau)}(x, \tau, m = 0) = \frac{\tau}{\pi(x^2 + \tau^2)} + \frac{1}{2}\delta(x - i\tau) + \frac{1}{2}\delta(x + i\tau), \quad (\text{F3})$$

i.e., when x is real, only the first term remains, which is the Cauchy distribution. Using the identity

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{\epsilon}{x^2 + \epsilon^2} = \delta(x), \quad (\text{F4})$$

we find that

$$\begin{aligned} \nabla^2 G^{(\tau)}(x, \tau, m = 0) &= -\frac{8\tau}{x^2 + \tau^2} \lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{\epsilon^2}{(x^2 + \tau^2 + \epsilon^2)^2} \\ &= -4f(x, \tau)\delta(x)\delta(\tau), \end{aligned} \quad (\text{F5})$$

where

$$\begin{aligned} f(x, \tau) &\equiv \frac{2\tau}{(x^2 + \tau^2)} = \frac{1}{\tau + ix} + \frac{1}{\tau - ix} \\ &= 2\pi G^{(\tau)}(x, \tau, m = 0). \end{aligned} \quad (\text{F6})$$

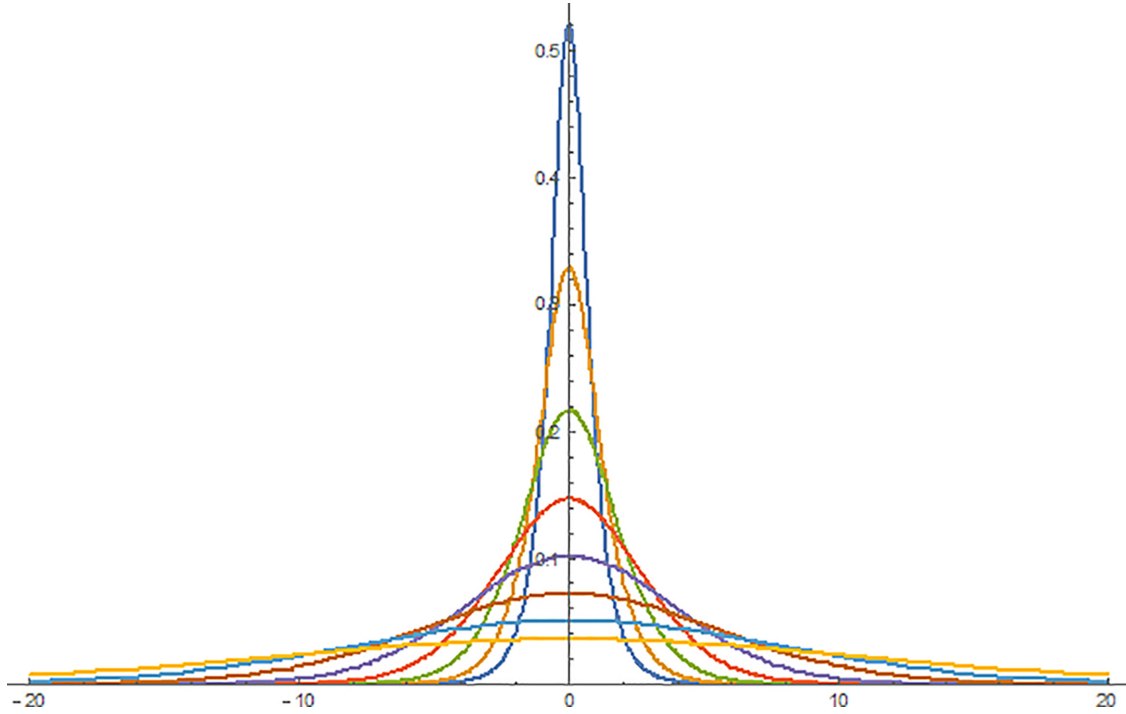
We immediately conclude that the same happens for general m ,

$$(\nabla^2 - m^2)G^{(\tau)}(x, \tau, m) = -4f(x, \tau)\delta(x)\delta(\tau). \quad (\text{F7})$$

Defining $r \equiv \sqrt{x^2 + \tau^2}$ and $\theta \equiv \arctan \frac{x}{\tau}$, we rewrite it as ($r, \theta \neq 0$)

$$G_{rr}^{(\tau)} + \frac{1}{r} G_r^{(\tau)} + \frac{1}{r^2} G_{\theta\theta}^{(\tau)} - m^2 G^{(\tau)} = 0, \quad (\text{F8})$$

where $G_{rr}^{(\tau)}$ means double derivative with respect to r , etc. Then, using the method of separation of variables

FIG. 6. A plot for Eq. (E28) for $\tau = 2^n$, $n = 0, 1, 2, 3, 4, 5, 6, 7$ (from top to bottom).

$G^{(\tau)} \equiv R(r)\Theta(\theta)$, we find

$$\begin{aligned} \Theta'' + n^2\Theta &= 0 \\ r^2 R'' + rR' - ((mr)^2 + n^2)R &= 0, \end{aligned} \quad (\text{F9})$$

from which one finds

$$\Theta = a_n \sin(n\theta) + b_n \cos(n\theta), \quad (\text{F10})$$

and R is a modified Bessel function of the second kind,

$$R(r) = K_n(mr) = \int_0^\infty \exp[-mr \cosh t] \cosh(nt) dt. \quad (\text{F11})$$

Note that we have two independent radial solutions, and the only one that is infinity at $r = 0$ and regular at $r \rightarrow \infty$ is this term. We find eventually

$$G^{(\tau)} = \frac{1}{2\pi} \sum_n K_n(mr) [a_n \sin(n\theta) + b_n \cos(n\theta)], \quad (\text{F12})$$

where a_n and b_n should be found using Eq. (F3) for $m = 0$. Given that,

$$K_n(y) \rightarrow \sqrt{\frac{\pi}{2}} \frac{\exp(-y)}{\sqrt{y}}, \quad y \rightarrow \infty,$$

$$K_0(y) \rightarrow -\gamma - \ln \frac{y}{2} + O(y^2), \quad y \rightarrow 0,$$

$$K_1(y) \rightarrow \frac{1}{y} + \frac{y}{2} \ln \frac{y}{2} + O(y), \quad y \rightarrow 0,$$

$$K_n(y) \rightarrow (-1)^{n-1} \left(\frac{y}{2} \right)^n \ln \frac{y}{2} \left(\frac{1}{n!} + O(y^2) \right), \quad n \geq 2, y \rightarrow 0. \quad (\text{F13})$$

Comparing this result with Eq. (E24), we notice that the singularity of the integral for $r \rightarrow 0$ (equivalent to $x = \pm t$)

is by $1/r$ and $r \ln r$, so that the only term that contributes in the summation is $n = 1$. More precisely, Eq. (E24) gives us

$$\begin{aligned} G &\rightarrow \frac{m}{2} \left[\frac{1}{mr} e^{-i\theta} + \frac{1}{2} \frac{mre^{i\theta}}{2} \ln(mr) \right] \\ &+ \frac{m}{2} \left[\frac{1}{mr} e^{i\theta} + \frac{1}{2} \frac{mre^{-i\theta}}{2} \ln(mr) \right] \\ &= m \left[\frac{1}{mr} + \frac{1}{2} \frac{mr}{2} \ln(mr) \right] \cos \theta. \end{aligned} \quad (\text{F14})$$

Then, we find that $a_1 = m$ and $b_1 = 0$, and all the other a_n, b_n ($n \neq 1$) are zero. The following integral identity confirms the above result:

$$\begin{aligned} \int_0^\infty dx \exp(-\alpha \sqrt{x^2 + \beta^2}) \cos \gamma x \\ = \frac{\alpha \beta}{\sqrt{\alpha^2 + \gamma^2}} K_1(\beta \sqrt{\alpha^2 + \gamma^2}), \quad \text{Re}[\alpha] > 0, \text{Re}[\beta] > 0. \end{aligned} \quad (\text{F15})$$

Using this, we find that

$$\begin{aligned} G(x, t) &= \frac{imt e^{imt}}{\pi \sqrt{x^2 - t^2}} K_1(m \sqrt{x^2 - t^2}), \quad x^2 > t^2 \\ &= -\frac{mt e^{imt}}{2 \sqrt{t^2 - x^2}} H_1^{(1)}(m \sqrt{t^2 - x^2}), \quad x^2 < t^2, \end{aligned} \quad (\text{F16})$$

where $H_1^{(1)}$ is a Hankel function of the first kind. Using new complex variables $z \equiv \tau + ix$ and $\bar{z} \equiv \tau - ix$, we can simplify the notation. First we note that $4\partial\bar{\partial} = \partial_\tau^2 + \partial_x^2$, where $\partial \equiv \partial_z$ and $\bar{\partial} \equiv \partial_{\bar{z}}$. Equation (F7) gives rise to

$$(4\partial\bar{\partial} - m^2)G^{(\tau)}(z, \bar{z}, m) = -2 \left(\frac{1}{z} + \frac{1}{\bar{z}} \right) \delta^2(z), \quad (\text{F17})$$

where $\delta^2(z) \equiv \delta(x)\delta(t)$, and we have used $\partial_{\bar{z}} \frac{1}{z} = \bar{\partial}_{\bar{z}} \frac{1}{z} = \delta^2(z)$. For the $m = 0$ case, $G^{(\tau)}$ is a summation of the holomorphic and antiholomorphic terms, i.e.,

$$G^{(\tau)}(z, \bar{z}, m = 0) = G_1(z) + G_2(\bar{z}) = \frac{1}{\pi} \left[\frac{1}{z} + \frac{1}{\bar{z}} \right]. \quad (\text{F18})$$

When $m \neq 0$, however, it is not decomposable into these two parts. In this case, Eq. (F15) tells us that

$$G^{(\tau)}(z, \bar{z}, m) \propto \frac{1}{\sqrt{z\bar{z}}} K_1(m\sqrt{z\bar{z}}). \quad (\text{F19})$$

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